

MY INTERNATIONAL PLACEMENT BY JAMES HANNON, UNIVERSITY COLLEGE DUBLIN

From February to June last year I undertook an international research placement at Queen's University Belfast. Under the supervision of **Professor Declan French** and **Dr Barry Quinn** I participated in a multidisciplinary project at the intersection of actuarial science, criminology and computer vision.



Our research focused on identifying environmental indicators of vehicle crime using street-level imagery. In standard insurance modelling, postcode data and historical claims are used to assess geographical risk. However, these proxies often fail to explain the underlying causes of the crime or reflect rapid changes in the urban landscape. To address this, we applied GroudingDINO, an open-set object detection algorithm to approximately two million Google Street View images across Northern Ireland. Guided by criminological literature, we detected signs of urban decay, such as litter, broken windows, and graffiti as well as deterrents like police stations, shopfronts, and CCTV cameras. We also identified flags and murals linked to self-policing communities and paramilitary influence.



The July 12th parade

This image-derived data was combined with vehicle crime records from the Police Service of Northern Ireland (PSNI), and spatial and regression models were used to assess the relationship between visual cues and crime levels. Controlling for socioeconomic variables using census data, our findings demonstrated the potential of computer vision to enhance the accuracy and fairness of insurance pricing models by incorporating dynamic and interpretable environmental features.

I had the opportunity to present this work both as a poster at the Conference on Applied Statistics in Ireland (CASI) and as a talk at the European Actuarial Journal Conference (EAJC). These forums allowed me to share findings with academic and professional audiences and to receive valuable feedback on the implications of this research for actuarial modelling and insurance practice.

Beyond the academic work, the placement provided a memorable cultural experience in Belfast, a city with a complex past and a hopeful future. I visited Crumlin Road Gaol, walked the Peace Bridge in Derry, and attended a July 12th parade to better understand local traditions. I also joined a boxing club in Sandy Row, a working-class loyalist area - something that would have been unthinkable during the Troubles, and a sign of how much the city has changed. Belfast's nightlife, sports scene and welcoming atmosphere made it an exceptional place to live and learn. This placement was a highlight of my PhD - academically enriching, personally meaningful and profoundly rewarding.



Crumlin Gaol in Belfast



The Peace Bridge in Derry